

Feature Engine Pro

Automated Feature Selection - Audit Report

Executive Summary

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34

INPUT FEATURES

2

SELECTED FEATURES

32

FEATURES DROPPED

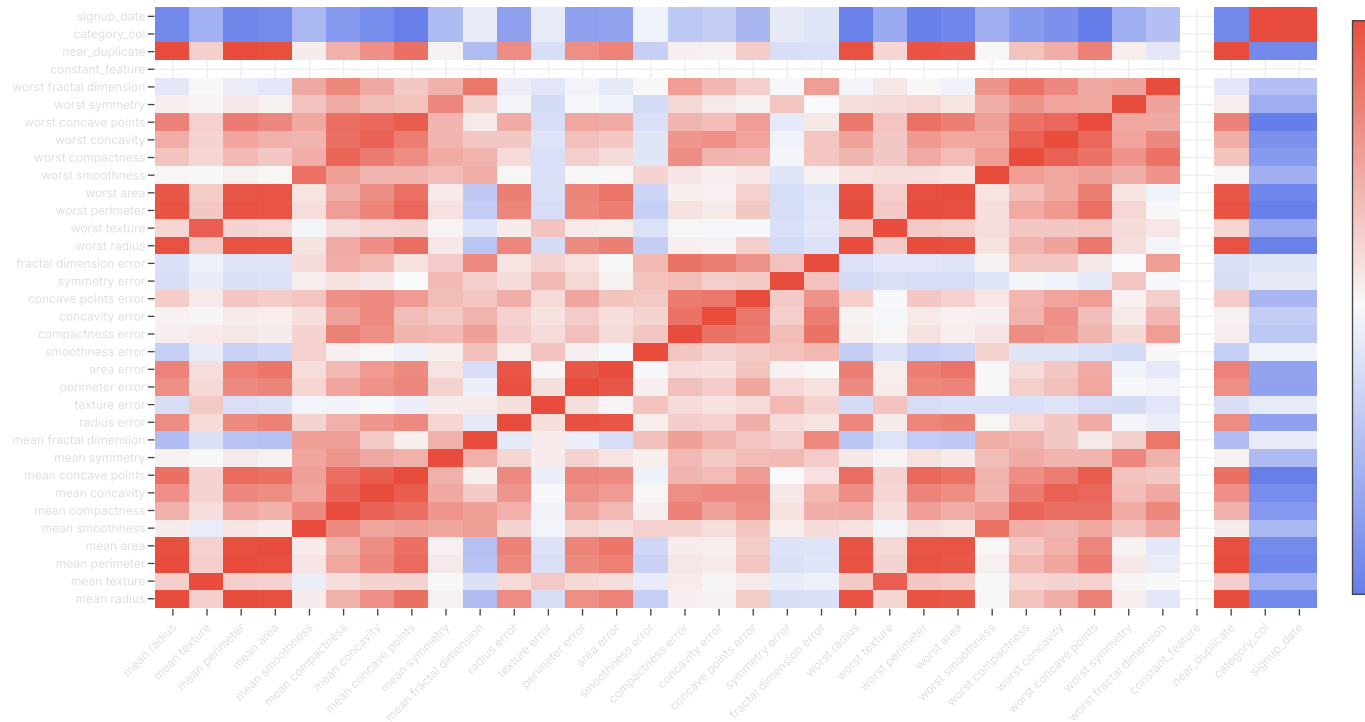
94.1%

DIMENSIONALITY
REDUCTION



Correlation Matrix

Interactive heatmap — hover for details, scroll to zoom



The Audit Trail

Complete history of feature transformations and selection reasoning

Search features, steps, or reasons...

- All Features (34)
- Kept (2)
- Dropped (32)

Feature	Final Status	Pipeline Step	Action & Reasoning
mean radius	✓ KEPT	VarianceThreshold	Variance 12.5020 is above threshold 0.01
mean texture	✓ KEPT	VarianceThreshold	Variance 17.9199 is above threshold 0.01
mean perimeter	✓ KEPT	VarianceThreshold	Variance 591.5610 is above threshold 0.01
mean area	✓ KEPT	VarianceThreshold	Variance 125984.6661 is above threshold 0.01
mean smoothness	✗ DROPPED	VarianceThreshold	Variance 0.0002 is below threshold 0.01
mean compactness	✗ DROPPED	VarianceThreshold	Variance 0.0028 is below threshold 0.01
mean concavity	✗ DROPPED	VarianceThreshold	Variance 0.0063 is below threshold 0.01
mean concave points	✗ DROPPED	VarianceThreshold	Variance 0.0014 is below threshold 0.01
mean symmetry	✗ DROPPED	VarianceThreshold	Variance 0.0008 is below threshold 0.01
mean fractal dimension	✗ DROPPED	VarianceThreshold	Variance 0.0001 is below threshold 0.01
radius error	✓ KEPT	VarianceThreshold	Variance 0.0802 is above threshold 0.01
texture error	✓ KEPT	VarianceThreshold	Variance 0.2935 is above threshold 0.01
perimeter error	✓ KEPT	VarianceThreshold	Variance 4.2899 is above threshold 0.01
area error	✓ KEPT	VarianceThreshold	Variance 2231.2698 is above threshold 0.01
smoothness error	✗ DROPPED	VarianceThreshold	Variance 0.0000 is below threshold 0.01
compactness error	✗ DROPPED	VarianceThreshold	Variance 0.0003 is below threshold 0.01
concavity error	✗ DROPPED	VarianceThreshold	Variance 0.0010 is below threshold 0.01
concave points error	✗ DROPPED	VarianceThreshold	Variance 0.0000 is below threshold 0.01
symmetry error	✗ DROPPED	VarianceThreshold	Variance 0.0001 is below threshold 0.01
fractal dimension error	✗ DROPPED	VarianceThreshold	Variance 0.0000 is below threshold 0.01
worst radius	✓ KEPT	VarianceThreshold	Variance 23.1483 is above threshold 0.01

Feature	Final Status	Pipeline Step	Action & Reasoning
worst texture	✓ KEPT	VarianceThreshold	Variance 36.7855 is above threshold 0.01
worst perimeter	✓ KEPT	VarianceThreshold	Variance 1113.8682 is above threshold 0.01
worst area	✓ KEPT	VarianceThreshold	Variance 322252.4539 is above threshold 0.01
worst smoothness	✗ DROPPED	VarianceThreshold	Variance 0.0005 is below threshold 0.01
worst compactness	✓ KEPT	VarianceThreshold	Variance 0.0240 is above threshold 0.01
worst concavity	✓ KEPT	VarianceThreshold	Variance 0.0438 is above threshold 0.01
worst concave points	✗ DROPPED	VarianceThreshold	Variance 0.0043 is below threshold 0.01
worst symmetry	✗ DROPPED	VarianceThreshold	Variance 0.0040 is below threshold 0.01
worst fractal dimension	✗ DROPPED	VarianceThreshold	Variance 0.0003 is below threshold 0.01
constant_feature	✗ DROPPED	VarianceThreshold	Variance 0.0000 is below threshold 0.01
near_duplicate	✓ KEPT	VarianceThreshold	Variance 12.5045 is above threshold 0.01
category_col	✓ KEPT	VarianceThreshold	Variance 0.2340 is above threshold 0.01
signup_date	✓ KEPT	VarianceThreshold	Variance 0.2340 is above threshold 0.01
mean radius	✗ DROPPED	CorrelationSelector	Correlated 1.00 with mean perimeter. Kept mean perimeter because it has higher correlation to target.
mean area	✗ DROPPED	CorrelationSelector	Correlated 0.99 with mean perimeter. Kept mean perimeter because it has higher correlation to target.
perimeter error	✗ DROPPED	CorrelationSelector	Correlated 0.97 with radius error. Kept radius error because it has higher correlation to target.
area error	✗ DROPPED	CorrelationSelector	Correlated 0.95 with radius error. Kept radius error because it has higher correlation to target.
mean perimeter	✗ DROPPED	CorrelationSelector	Correlated 0.97 with worst radius. Kept worst radius because it has higher correlation to target.
mean texture	✗ DROPPED	CorrelationSelector	Correlated 0.90 with worst texture. Kept worst texture because it has higher correlation to target.
worst radius	✗ DROPPED	CorrelationSelector	Correlated 0.99 with worst perimeter. Kept worst perimeter because it has higher correlation to target.

Feature	Final Status	Pipeline Step	Action & Reasoning
worst area	X DROPPED	CorrelationSelector	Correlated 0.98 with worst perimeter. Kept worst perimeter because it has higher correlation to target.
worst compactness	X DROPPED	CorrelationSelector	Correlated 0.89 with worst concavity. Kept worst concavity because it has higher correlation to target.
near_duplicate	X DROPPED	CorrelationSelector	Correlated 0.96 with worst perimeter. Kept worst perimeter because it has higher correlation to target.
signup_date	X DROPPED	CorrelationSelector	Correlated 1.00 with category_col. Kept category_col because it has higher correlation to target.
radius error	✓ KEPT	CorrelationSelector	No collinearity above 0.85 threshold found
texture error	✓ KEPT	CorrelationSelector	No collinearity above 0.85 threshold found
worst texture	✓ KEPT	CorrelationSelector	No collinearity above 0.85 threshold found
worst perimeter	✓ KEPT	CorrelationSelector	No collinearity above 0.85 threshold found
worst concavity	✓ KEPT	CorrelationSelector	No collinearity above 0.85 threshold found
category_col	✓ KEPT	CorrelationSelector	No collinearity above 0.85 threshold found
radius error	✓ KEPT	MutualInformation	Mutual Information 0.2471 >= threshold 0.01.
texture error	X DROPPED	MutualInformation	Mutual Information 0.0011 < threshold 0.01. Lacks non-linear dependency.
worst texture	✓ KEPT	MutualInformation	Mutual Information 0.1473 >= threshold 0.01.
worst perimeter	✓ KEPT	MutualInformation	Mutual Information 0.4621 >= threshold 0.01.
worst concavity	✓ KEPT	MutualInformation	Mutual Information 0.3148 >= threshold 0.01.
category_col	✓ KEPT	MutualInformation	Mutual Information 0.6608 >= threshold 0.01.
radius error	X DROPPED	RFE	RFE Ranked 3. Eliminated due to low tree-based feature importance.
worst texture	X DROPPED	RFE	RFE Ranked 4. Eliminated due to low tree-based feature importance.
worst perimeter	✓ KEPT	RFE	RFE Ranked 1 (Top Tier Feature Importance).
worst concavity	X DROPPED	RFE	RFE Ranked 2. Eliminated due to low tree-based feature importance.

Feature	Final Status	Pipeline Step	Action & Reasoning
category_col	✓ KEPT	RFE	RFE Ranked 1 (Top Tier Feature Importance).